

Reports

1. Feature Overview

Staff **Reports** hub with Chart.js UI and OpenPDF downloads:

- Transaction trends (credits / debits over a date range)
- Account mix (counts by account type / status)
- Transfer approvals summary (ADMIN / MANAGER / AUDITOR)

Status: Implemented

2. Business Purpose

Management visibility without raw SQL: trends, portfolio mix, approval workload.

3. User Workflow

1. Sidebar → **Reports** (roles with `ACCOUNTS_READ` ; approvals category restricted to Admin/Manager/Auditor).
2. Choose category → set filters → view chart + on-screen table.
3. Download PDF.

4. Execution Flow



5. Database Tables

No dedicated report tables. Reads `bank_transaction` , `account` , `transfer_request` .

6. REST APIs

Base: `/reports`

Method	Path	Access
GET GET	<code>/reports/transaction-trends</code>	<code>ACCESS_ACCOUNTS_READ</code> same
GET GET	<code>/reports/transaction-trends/pdf</code>	<code>ACCESS_ACCOUNTS_READ</code> same ADMIN
GET GET	<code>/reports/account-mix</code> <code>/reports/account-mix/pdf</code> <code>/reports/approvals</code>	MANAGER AUDITOR same
	<code>/reports/approvals/pdf</code>	

Query params typically include `from / to` (ISO dates) where applicable. No hard 93-day cap (removed for learning flexibility).

7–13. Code map

Layer	Classes / paths
Backend	<code>com.bankone.report.*</code> — <code>ReportController</code> , services, DTOs, PDF helpers
Frontend	<code>features/reports/reports.*</code> — <code>Chart.js</code> , category picker

14–18. Validation / Security / Logging / Audit

- Approvals reports: role-gated in `SecurityConfig`.
- Large ranges may be slow on load-test DBs — candidate for caching / read-replica learning topics.

19. Testing Strategy

Compare trend totals to known deposit fixtures; open PDF in browser.

20. Future Extension Guide

Cursor aggregation jobs; cache daily buckets; see `TECH_LEARNING_PLAN`.

Future Modification Guide

Requirement: Add branch filter to account mix

Item	Detail
Files Impact	Report DTO + service query; Angular filter control Needs consistent <code>branch_code</code> data quality